

Kubernetes DNS Resolution Issue SOP

Objective

Restore service discovery when Kubernetes service names or internal DNS records do not resolve from application pods.

Scope

Applicable to Kubernetes workloads, platform operations, DevOps teams, SRE teams, application teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

- 1. Access to Kubernetes cluster using kubectl and the correct namespace context.
- 2. Access to application, pod, deployment, service, ingress, and node logs where applicable.
- 3. Access to monitoring dashboards such as Grafana, Prometheus, cloud monitoring, or container insights.
- 4. Permission to inspect manifests, secrets, configmaps, RBAC bindings, PVCs, and rollout history.
- 5. Escalation contact for DevOps, platform, infrastructure, network, and application teams.

Incident Metadata

Field	Value
Ticket ID	T035
Issue	Kubernetes DNS resolution issue
Category	Networking
Service	Kubernetes
Severity	Medium
Status	Resolved
Description	Service names not resolving
Expected Resolution	Check CoreDNS configuration

Procedure

- 1. Confirm failing DNS name, namespace, pod, and exact lookup error.
- 2. Test DNS from a debug pod using nslookup or dig.
- 3. Check CoreDNS pod status, restarts, logs, and ConfigMap.
- 4. Validate service and endpoints exist for the DNS name being queried.
- 5. Check pod DNS policy, resolv.conf, search domains, and network policy.
- 6. Review recent CoreDNS, CNI, or cluster network changes.
- 7. Restart CoreDNS only if it is unhealthy and approved.
- 8. Retest DNS lookup and application connectivity.

Common Commands

```
kubectl config current-context
kubectl get pods -n <namespace> -o wide
kubectl describe pod <pod-name> -n <namespace>
kubectl logs <pod-name> -n <namespace> --tail=100
kubectl get events -n <namespace> --sort-by=.lastTimestamp
kubectl get deploy,svc,ingress,pvc -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Service name not resolving	CoreDNS unhealthy	Check CoreDNS pods and logs
Only one namespace affected	Wrong DNS name or search domain	Use full service DNS name
DNS resolves but app fails	Service endpoints missing	Fix backend service/pods
Timeout on lookup	Network policy or CNI issue	Check DNS traffic policy

Best Practices

- 1. Monitor CoreDNS latency and errors.
- 2. Use fully qualified service names for cross-namespace calls.
- 3. Avoid unnecessary CoreDNS ConfigMap edits.
- 4. Keep DNS troubleshooting commands in runbooks.

References

[Kubernetes Docs](#) | [Internal Tickets CSV](#) | [Cluster Events](#) | [Monitoring Dashboards](#) | [Application Logs](#) | [Platform Runbook](#)